

CMPG 325 COMPUTER NETWORKS

Individual Semester Project Milestone 2: Client Implementation Review

Student: Nemaguvhuni, P

Student Number: 40963160

Project ID: CMPG325-2026-102

Client ID: CLI-102

Assigned Organisation: Greater Taung Local Municipality Offices (Taung)

Industry: Municipal Services

1. INTRODUCTION

This project involves the design and implementation of a computer network for the Greater Taung Local Municipality Offices using Cisco Packet Tracer. The assigned networking challenge is Basic OSPF Single-Area Dynamic Routing.

The project also includes Change Request 3 (CR3), which requires a guest Wi-Fi network for visitors that is isolated from the internal municipal network.

The network uses the assigned address block 172.30.70.0/23 and includes routers, switches, internal end devices, a municipal server and a guest wireless network.

This document presents the implemented networking features, configuration evidence, connectivity testing, guest network isolation testing and important troubleshooting performed during the implementation.

2. ASSIGNED FEATURE IMPLEMENTATION

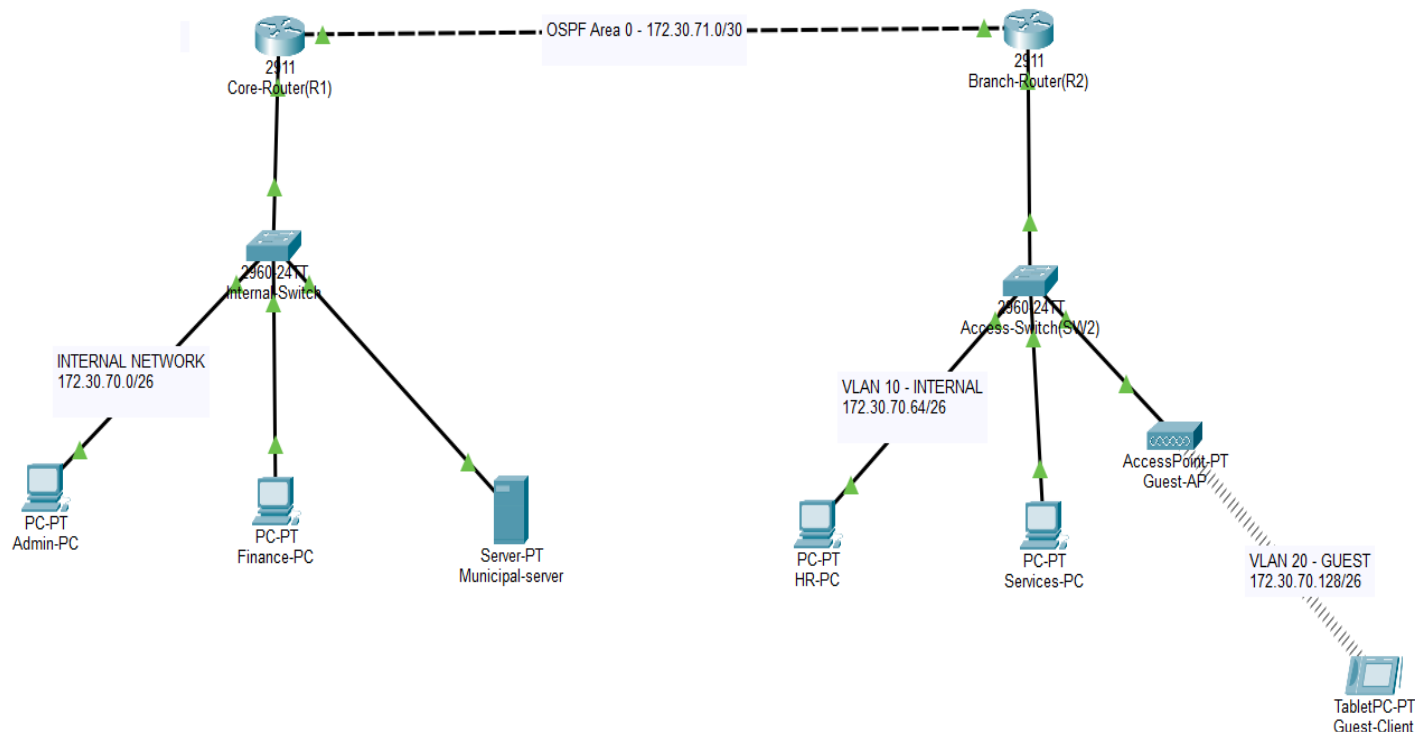


Figure 1: Cisco Packet Tracer network implementation

2.1 BASIC OSPF SINGLE AREA DYNAMIC ROUTING

The assigned networking challenge was implemented using Open Shortest Path First (OSPF) between the Core Router (R1) and Branch Router (R2).

OSPF was configured using Area 0. The connection between R1 and R2 uses the network 172.30.71.0/30.

R1 advertises the internal LAN network 172.30.70.0/26, while R2 advertises the internal VLAN 10 network 172.30.70.64/26.

The guest VLAN 20 network, 172.30.70.128/26, was not included in the OSPF network statements. Guest isolation is handled separately using an ACL on R2.

OSPF Neighbour Verification

The OSPF neighbour relationship was checked on both routers using the **show ip ospf neighbor** command.

R1 shows R2 in the FULL/DR state, and R2 shows R1 in the FULL/BDR state. This confirms that the OSPF neighbour relationship was successfully established.

```
Router>enable
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.30.71.1	1	FULL/BDR	00:00:35	172.30.71.1	GigabitEthernet0/0

```
Router#
```

Figure 2: R2 OSPF neighbour verification

```
Router>enable
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.30.71.2	1	FULL/DR	00:00:34	172.30.71.2	GigabitEthernet0/0

```
Router#
```

Figure 3: R1 OSPF neighbour verification

OSPF Routing Verification

The routing table on R2 was checked using **show ip route**.

The output shows:

O 172.30.70.0/26 [110/2] via 172.30.71.1

The O indicates that the route was learned through OSPF. This confirms that R2 is dynamically learning the R1 internal LAN network.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

172.30.0.0/16 is variably subnetted, 7 subnets, 3 masks
O       172.30.70.0/26 [110/2] via 172.30.71.1, 00:29:10, GigabitEthernet0/0
C       172.30.70.64/26 is directly connected, GigabitEthernet0/1.10
L       172.30.70.65/32 is directly connected, GigabitEthernet0/1.10
C       172.30.70.128/26 is directly connected, GigabitEthernet0/1.20
L       172.30.70.129/32 is directly connected, GigabitEthernet0/1.20
C       172.30.71.0/30 is directly connected, GigabitEthernet0/0
L       172.30.71.2/32 is directly connected, GigabitEthernet0/0

Router#
```

Figure 4: R2 routing table showing the OSPF-learned route

2.2 GUEST WIFI AND ISOLATION

Change Request 3 was implemented by creating a separate Guest VLAN 20 for visitors. VLAN 20 uses the network 172.30.70.128/26, with 172.30.70.129 configured as the gateway. The Guest-Client uses 172.30.70.130.

The guest network is connected through the wireless access point and VLAN 20 is carried across the switch trunk.

VLAN Configuration

The VLAN configuration was verified using **show vlan brief**.

The output confirms that VLAN 10, named INTERNAL, and VLAN 20, named GUEST, are active.

```

Switch>enable
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   INTERNAL              active    Fa0/2, Fa0/3
20   GUEST                 active    Fa0/4
1002 fddi-default         active
1003 token-ring-default   active
1004 fddinet-default      active
1005 trnet-default        active
Switch#

```

Figure 5: Access Switch (SW2) VLAN configuration

Trunk Configuration

The connection between SW2 and R2 was verified using **show interfaces trunk**.

The output confirms that the trunk uses 802.1Q and that VLANs 10 and 20 are active and forwarding.

```

1005 trnet-default                active
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20

Switch#

```

Figure 6: Access-Switch (SW2) trunk configuration

GUEST ISOLATION ACL

An extended ACL was configured on R2's G0/1.20 sub interface to prevent guest traffic from reaching the internal networks.

The ACL blocks traffic from 172.30.70.128/26 to the internal range 172.30.70.0/25 while allowing other traffic.

The ACL was verified using **show access-lists 100**. The deny statement recorded matches, confirming that guest-to-internal traffic was being blocked.

```
Router#show access-lists 100
Extended IP access list 100
  deny ip 172.30.70.128 0.0.0.63 172.30.70.0 0.0.0.127 (8 match(es))
  permit ip any any (4 match(es))

Router#
```

Copy

Figure 7: ACL 100 verification

3. TESTING EVIDENCE

Testing was performed to confirm that the network was functioning correctly and that the implemented features behaved as intended.

3.1. INTERNAL NETWORK CONNECTIVITY

Admin-PC to HR-PC

A ping was performed from Admin-PC to HR-PC at 172.30.70.66.

The test returned 4/4 successful replies with 0% packet loss, confirming connectivity between the internal LAN and VLAN 10.

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.70.66

Pinging 172.30.70.66 with 32 bytes of data:

Request timed out.
Reply from 172.30.70.66: bytes=32 time=11ms TTL=126
Reply from 172.30.70.66: bytes=32 time=2ms TTL=126
Reply from 172.30.70.66: bytes=32 time=19ms TTL=126

Ping statistics for 172.30.70.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 19ms, Average = 10ms

C:\>ping 172.30.70.66

Pinging 172.30.70.66 with 32 bytes of data:

Reply from 172.30.70.66: bytes=32 time<1ms TTL=126
Reply from 172.30.70.66: bytes=32 time=1ms TTL=126
Reply from 172.30.70.66: bytes=32 time<1ms TTL=126
Reply from 172.30.70.66: bytes=32 time<1ms TTL=126

Ping statistics for 172.30.70.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 8: Admin-PC to HR-PC connectivity test

Admin-PC to Municipal-Server

A ping was performed from Admin-PC to the Municipal-Server at 172.30.70.10.

The test returned 4/4 successful replies with 0% packet loss, confirming network-level connectivity between the internal workstation and the municipal server.

The Municipal-Server represents the server used for the municipal file, print and application services required by the project. This test confirms network connectivity to the server.

```
C:\>ping 172.30.70.10

Pinging 172.30.70.10 with 32 bytes of data:

Reply from 172.30.70.10: bytes=32 time<1ms TTL=128
Reply from 172.30.70.10: bytes=32 time<1ms TTL=128
Reply from 172.30.70.10: bytes=32 time<1ms TTL=128
Reply from 172.30.70.10: bytes=32 time<1ms TTL=128

Ping statistics for 172.30.70.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 9: Admin-PC to Municipal-Server connectivity test

HR-PC to Municipal-Server

The project requires the critical municipal services to remain available during business hours. To test connectivity to the server from another part of the network, a ping was performed from HR-PC to the Municipal-Server at 172.30.70.10.

The test returned 4/4 successful replies with 0% packet loss, confirming that the Municipal-Server is reachable from the R2-side internal network through the OSPF connection. This demonstrates that users on this side of the network can reach the server hosting the critical municipal services.

```
C:\>ping 172.30.70.10

Pinging 172.30.70.10 with 32 bytes of data:

Reply from 172.30.70.10: bytes=32 time<1ms TTL=126
Reply from 172.30.70.10: bytes=32 time<1ms TTL=126
Reply from 172.30.70.10: bytes=32 time=11ms TTL=126
Reply from 172.30.70.10: bytes=32 time<1ms TTL=126

Ping statistics for 172.30.70.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>
```

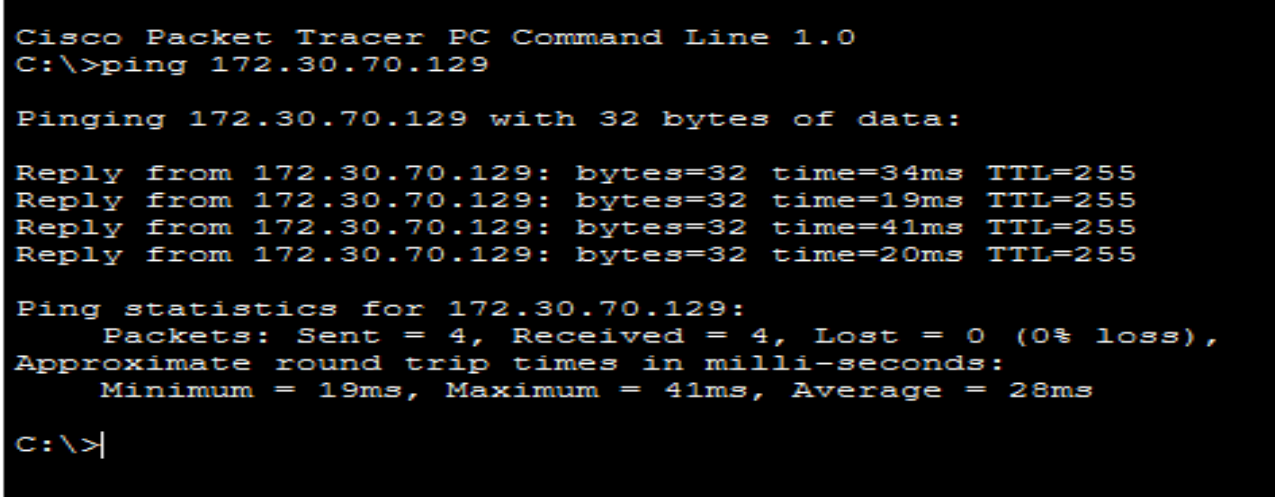
Figure 10: HR-PC to Municipal-Server connectivity test

3.2 GUEST NETWORK CONNECTIVITY

Guest-Client to Guest Gateway

The Guest-Client was tested against its gateway at 172.30.70.129.

The test returned 4/4 successful replies with 0% packet loss, confirming that the Guest-Client had working connectivity within the guest network.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.70.129

Pinging 172.30.70.129 with 32 bytes of data:

Reply from 172.30.70.129: bytes=32 time=34ms TTL=255
Reply from 172.30.70.129: bytes=32 time=19ms TTL=255
Reply from 172.30.70.129: bytes=32 time=41ms TTL=255
Reply from 172.30.70.129: bytes=32 time=20ms TTL=255

Ping statistics for 172.30.70.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 41ms, Average = 28ms

C:\>|
```

Figure 11: Guest-Client to guest gateway connectivity test

3.3 GUEST NETWORK ISOLATION

Guest network isolation was tested by attempting to reach internal municipal resources from the Guest-Client.

Guest-Client to HR-PC

Before the ACL was implemented, the Guest-Client was able to reach HR-PC. This showed that the guest network was connected but was not yet isolated from the internal network.

After ACL 100 was applied, the test was repeated.

The Guest-Client was unable to reach HR-PC, resulting in 0/4 successful replies and 100% packet loss. The response showed: Reply from 172.30.70.129: Destination host unreachable.

This confirms that guest-to-internal traffic was being blocked.

```
C:\>ping 172.30.70.66

Pinging 172.30.70.66 with 32 bytes of data:

Reply from 172.30.70.129: Destination host unreachable.
Reply from 172.30.70.129: Destination host unreachable.
Reply from 172.30.70.129: Destination host unreachable.
Reply from 172.30.70.129: Destination host unreachable.

Ping statistics for 172.30.70.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 12: Guest-Client to HR-PC isolation test

Guest-Client to Municipal-Server

The Guest-Client was also tested against the Municipal-Server at 172.30.70.10.

The test resulted in 0/4 successful replies and 100% packet loss, with the guest gateway returning a destination unreachable message. This confirms that the guest network cannot reach the municipal server.

```
C:\>ping 172.30.70.10

Pinging 172.30.70.10 with 32 bytes of data:

Reply from 172.30.70.129: Destination host unreachable.
Reply from 172.30.70.129: Destination host unreachable.
Reply from 172.30.70.129: Destination host unreachable.
Reply from 172.30.70.129: Destination host unreachable.

Ping statistics for 172.30.70.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 13: Guest-Client to Municipal-Server isolation test

4. TROUBLESHOOTING

Several issues were identified and resolved during the implementation.

4.1. GUEST WIRELESS DEVICE

The original Guest-Client was configured as a Laptop-PT. The device did not have a Wireless0 interface available in the Packet Tracer version being used, and the wireless interface could not be installed.

The issue was resolved by replacing the Laptop-PT with a TabletPC-PT, which supported wireless connectivity.

4.2 GUEST-CLIENT CONFIGURATION

The replacement Guest-Client was configured with:

- IP address: 172.30.70.130
- Subnet mask: 255.255.255.192
- Default gateway: 172.30.70.129

The Guest-Client was then tested against its gateway and successfully received replies.

This is shown in Figure 11.

4.3. GUEST ISOLATION

After the Guest-Client was successfully connected, a test to HR-PC initially succeeded. This showed that the guest network was working but guest traffic could still reach the internal network.

An ACL was therefore configured on R2's(router) G0/1.20 sub interface to block traffic from the guest subnet to the internal networks.

The test was repeated after applying the ACL, and guest access to HR-PC and the Municipal-Server was successfully blocked.

The ACL was also checked using show access-lists 100, which showed matches on the deny statement.

This is shown in Figure 7, Figure 12, Figure 13.

5. TESTING SUMMARY

Test/Verification	Result	Evidence
R2 OSPF neighbour	FULL/BDR	Figure 2
R1 OSPF neighbour	FULL/DR	Figure 3
OSPF route on R2	R1 LAN learned through OSPF	Figure 4
Admin-PC to HR-PC	4/4, 0% loss	Figure 8
Admin-PC to Municipal-Server	4/4, 0% loss	Figure 9
HR-PC to Municipal-Server	4/4, 0% loss	Figure 10
Guest-Client to Gateway	4/4, 0% loss	Figure 11
Guest-Client to HR-PC	0/4, 100% loss	Figure 12
Guest-Client to Municipal-Server	0/4, 100% loss	Figure 13
ACL 100	Deny rule recorded matches	Figure 7
VLAN configuration	VLAN 10 and 20 active	Figure 5
Trunk configuration	VLANs 10 and 20 forwarding	Figure 6

6. CONCLUSION

The network was successfully implemented and tested in Cisco Packet Tracer. The assigned Basic OSPF Single-Area Dynamic Routing feature was configured between R1 and R2 and verified using OSPF neighbour information and the routing table.

Change Request 3 was implemented using a separate Guest VLAN 20 and an ACL to prevent guest users from accessing internal municipal resources.

Relevant end-to-end connectivity was tested, including internal connectivity and guest network connectivity. Guest isolation was also tested by attempting to reach internal resources from the Guest-Client.

Clear evidence was captured for the configuration and testing results, and the important troubleshooting performed during implementation was documented.

The final Packet Tracer file was reopened and verified to confirm that the working configuration was retained.

7. Updated GitHub portfolio

<https://github.com/Phathu647/CMPG325-Greater-Taung-Municipal-Network.git>